

# PRV

## Pressure relief valves

### Installation and operating instructions



Other languages

<http://net.grundfos.com/qr/i/96681525>

be  
think  
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**GRUNDFOS** 

# English (GB) Installation and operating instructions

## Original installation and operating instructions

These installation and operating instructions describe the Grundfos PRV (Pressure Relief Valve).

Sections 1-4 give the information necessary to be able to install and start up the product in a safe way.

Sections 5-10 give important information about the product as well as information on service, fault finding and disposal of the product.

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## 1. General information

### 1.1 Target group

This document is intended for the operating company and the users. It contains general instructions that must be observed before installation and during operation and maintenance of the product. The responsible staff must read these instructions prior to any work on the product.

#### 1.1.1 Qualification and training

The persons responsible for the tasks described in this document must be appropriately qualified.

#### 1.1.2 Obligations of the operating company

- Observe the local safety regulations.
- Keep the installation and operating instructions available at the installation location at all times.
- Coordinate the preparation of the installation location observing section 9. *Technical data*.
- Ensure that the users are trained for their tasks.
- Provide the stipulated safety equipment and personal protective equipment.
- Arrange regular maintenance.

#### 1.1.3 Obligations of the user

- Observe the recognised health and safety regulations as well as the local accident prevention regulations.
- Wear protective equipment in accordance with local health and safety regulations when working on the product and handling chemicals.
- Read and understand this document.

### 1.2 Safe operation

When working with chemicals, the accident prevention regulations applicable at the installation site must be applied.

Observe the chemical manufacturer's safety data sheets when handling chemicals.

When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.

Only operate the system if all lines are connected correctly.

If safe operation is no longer possible, the product must be taken out of operation and secured against unintentional operation.

This is the case in the following situations:

- If the product is visibly damaged.
- If the product does not seem operational.
- After long periods of storage under unfavourable conditions.



Read this document before installing the product. Installation and operation must comply with local regulations and accepted codes of good practice.

### 1.3 Symbols used in this document



#### WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.

## 2. Installing the product

### 2.1 Location

- The installation location must be protected from rain, humidity, condensation, direct sunlight and dust.
- The installation location must have sufficient lighting to ensure safe operation.
- Observe the permissible ambient conditions. See section [9.2 Storage and ambient temperatures](#).

### 2.2 Mechanical installation

The product must only be installed by authorised and qualified persons.

When working with chemicals, the accident prevention regulations applicable at the installation site must be applied.



Observe the chemical manufacturer's safety data sheets when handling chemicals.

When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.

Observe section [4.1 Handling the product](#).

Wipe up spilled liquid immediately to avoid slipping hazard.

#### 2.2.1 Installation requirements

Pressure relief valves are installed in the outlet line. They can also be mounted directly on the outlet valve of the dosing pump with an optional adapter.



If a pressure loading valve is used in conjunction with a pressure relief valve, the pressure loading valve must be installed downstream the pressure relief valve.

If a pressure relief valve is used conjunction with a pulsation damper DBG, the pressure relief valve must be installed upstream the pulsation damper.

### 2.2.2 Installation example

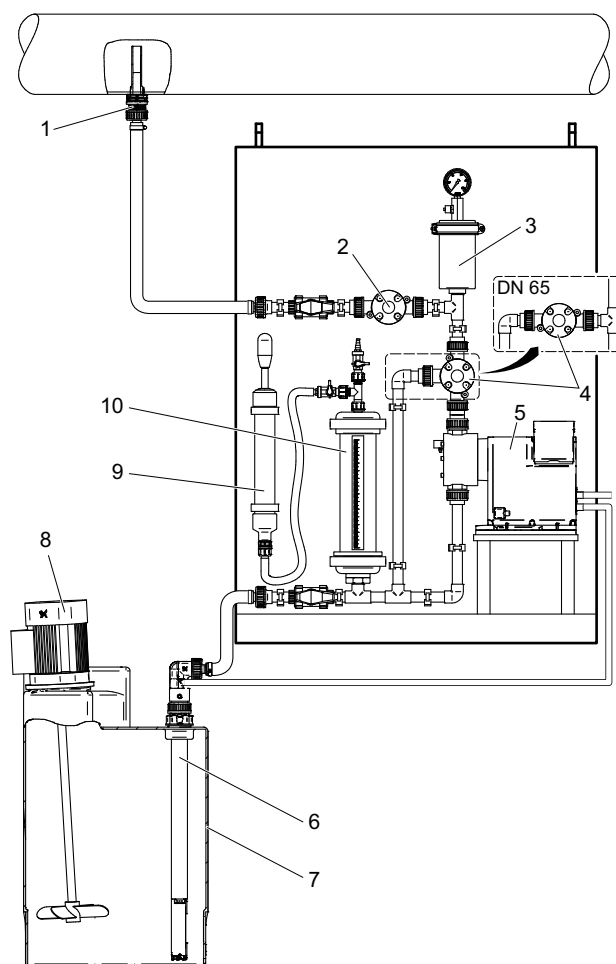


Fig. 1 Installation example

| Pos. | Description                |
|------|----------------------------|
| 1    | Injection unit             |
| 2    | Pressure loading valve PLV |
| 3    | Pulsation damper DBG       |
| 4    | Pressure relief valve PRV  |
| 5    | Dosing pump                |
| 6    | Rigid suction lance RSL    |
| 7    | Dosing tank                |
| 8    | Electric stirrer           |
| 9    | Manual vacuum pump         |
| 10   | Pulsation damper CSD       |

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### 2.2.3 Pressure relief valves up to 460 l/h

Pressure relief valves up to 460 l/h are designed as three-way valves. They have an inlet and an outlet on opposed sides, and an unpressurised overflow outlet. The flow direction is indicated by an arrow on the valve body.

The pressure relief valves can be installed in the outlet line, or mounted directly on the outlet valve of the dosing pump with an optional adapter, see section [7.1 Union nut adapters \(up to 460 l/h\)](#).

#### Installing the valve in the outlet line

Observe section [2.2.1 Installation requirements](#).



Install the valve in flow direction. The flow direction is indicated by an arrow on the valve body.

1. Install the valve in the outlet line.
2. Connect the overflow line to the overflow outlet, and direct the unpressurised liquid to the tank or a suitable overflow.
3. Re-tighten the screws on the top part of the valve after 48 operating hours with a torque wrench. Max. torque:

| Up to 60 l/h | 60-460 l/h | DN 65 |
|--------------|------------|-------|
| 2 Nm         | 4 Nm       | 5 Nm  |

#### Installing the valve directly on the pump

Valves up to 460 l/h can be installed directly on the pump with a special adapter. See section [7.1 Union nut adapters \(up to 460 l/h\)](#).

1. Remove the connector from the inlet of the valve.
2. Screw the adapter onto the inlet of the valve.
3. Screw the valve with the adapter onto the outlet valve of the pump.
4. Connect the outlet line to the outlet of the valve.
5. Connect the overflow line to the overflow outlet, and direct the unpressurised liquid to the tank or a suitable overflow.
6. Re-tighten the screws on the top part of the valve after 48 operating hours with a torque wrench. Max. torque:

| Up to 60 l/h | 60-460 l/h | DN 65 |
|--------------|------------|-------|
| 2 Nm         | 4 Nm       | 5 Nm  |

### 2.2.4 Pressure relief valves DN 65

Pressure relief valves DN 65 are usually designed as bypass pressure relief valves with connected T-piece for installation in the outlet line.

The relief pressure of the valve depends on the direction of installation, i.e. the direction the liquid flows through the valve (see fig. 2). The T-pieces are pre-assembled according to the selected relief pressure. See section [9. Technical data](#).

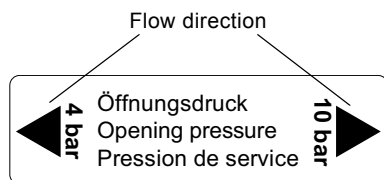


Fig. 2 Indication of relief pressure on valve body

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The T-piece is not always supplied as standard. There are some special versions in the form of three-way valves similar to the valves up to 60 l/h. Please always observe the corresponding dimensional drawings in section [9. Technical data](#)! For three-way valves proceed as described in section [2.2.3 Pressure relief valves up to 460 l/h](#).

#### Installing the valve

1. Install the T-piece with the valve in the outlet line.
2. Connect the overflow line to the pressure relief valve and direct the unpressurised liquid to the tank or a suitable overflow.

### 3. Commissioning the product

The product must only be adjusted and commissioned by authorised and qualified persons. When working with chemicals, the accident prevention regulations applicable at the installation site must be applied.



Observe the chemical manufacturer's safety data sheets when handling chemicals.

When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.

Wipe up spilled liquid immediately to avoid slipping hazard.

The relief pressure of the valve is set in the factory to the value specified in the technical data. See section [9. Technical data](#). During operation, the relief pressure depends on various factors, e.g. the flow, the stroke frequency of the pump, or the counterpressure. If a more precise setting is required, the valve must be adapted to the local conditions.

#### 3.1 Setting the relief pressure

If the dosing pump does not provide counterpressure measurement, a pressure gauge must be installed in the system between the pump and the pressure relief valve.

#### WARNING

##### Pressurised dosing medium

- Death or serious personal injury
- Set the relief pressure lower than the maximum permissible operating pressure of the dosing system.

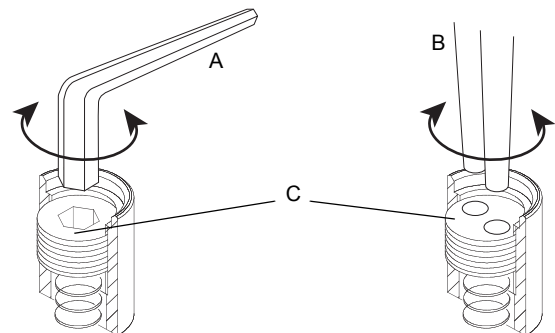


Fig. 3 Setting the relief pressure

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### 3.1.1 Optimal installation

The following steps apply to installations similar to the installation example. See fig. 1.

While the pump is running:

1. Remove the protective cap from the top part of the valve.
2. Close the isolating valve behind the pressure loading valve (2).
3. When you hear the dosing medium overflowing, read the current relief pressure from the pump or the pressure gauge.
4. For valves up to 460 l/h use a hexagon wrench (A). For DN 65 valves use a pair of needle-nosed pliers (B):
  - turn adjustment screw (C) clockwise to increase the pressure
  - turn adjustment screw (C) counterclockwise to reduce the pressure
5. Set the desired relief pressure.
6. Open the isolating valve behind the pressure loading valve (2).
7. Attach the protective cap again.

### 3.1.2 Other installations

While the pump is running:

1. Remove the protective cap from the top part of the valve.
2. Set the system pressure slightly below the desired maximum operating pressure.
3. For valves up to 460 l/h use a hexagon wrench (A). For DN 65 valves use a pair of needle-nosed pliers (B):
  - Turn adjustment screw (C) counterclockwise to reduce the relief pressure until medium flows over.
4. Turn adjustment screw (C) clockwise by half a turn. The relief pressure is now set slightly above the current operating pressure.
5. Attach the protective cap again.

## 4. Handling and storing the product

### 4.1 Handling the product

- Wear safety shoes during transport and installation.
- Wear a helmet if lifting the product higher than hip level.
- Use appropriate lifting and transporting equipment.
- Make sure that the product is not exposed to any point load during the transport.
- Avoid strong impacts.
- Observe the permissible ambient conditions. See section [9.2 Storage and ambient temperatures](#).

### 4.2 Storing the product

- Observe the permissible ambient conditions. See section [9.2 Storage and ambient temperatures](#).
- The storage location must be protected from rain, humidity, condensation, direct sunlight and dust.

## 5. Product introduction

### 5.1 Intended use

The pressure relief valve PRV protects the outlet line against impermissibly high pressure. If the pressure rises above the set relief pressure of the valve, the valve opens, and the dosing medium passing through the valve can flow back to the tank or to a suitable overflow.

The valve is suitable for liquid, non-abrasive, non-flammable and non-combustible media.

Observe the freezing and boiling points of the medium.

Make sure that parts in contact with the medium are resistant to the medium under operating conditions.



The product must not be used for any other purpose than the one mentioned above.

#### 5.1.1 Improper operating methods

The operational safety of the product is only guaranteed, if it is used in accordance with section [5.1 Intended use](#).

The product must not be used for:

- operation in potentially explosive areas
- frozen media or gases
- crystallising media.

## 5.2 Identification

### 5.2.1 Nameplate (pressure relief valves from 60 to 460 l/h)



Fig. 4 Nameplate

| Pos. | Description                 |
|------|-----------------------------|
| 1    | Product number              |
| 2    | Week and year of production |
| 3    | Type key                    |
| 4    | Nominal width               |
| 5    | Body material               |
| 6    | Gasket material             |
| 7    | Factory-set relief pressure |
| 8    | Maximum operating pressure  |
| 9    | Country of origin           |

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## 5.2.2 Type key

## Pressure relief valves up to 60 l/h

| Code                                | Description                                                             | PRV- | G5/8- | 10 | PP/ | V,E | U2 |
|-------------------------------------|-------------------------------------------------------------------------|------|-------|----|-----|-----|----|
| PRV                                 | Pressure relief valve                                                   |      |       |    |     |     |    |
| <b>Connection size</b>              |                                                                         |      |       |    |     |     |    |
| G5/8                                | G 5/8 external thread, groove for O-ring on top                         |      |       |    |     |     |    |
| G1/4                                | Rp 1/4 internal thread (only SS versions)                               |      |       |    |     |     |    |
| 1/4N                                | 1/4 NPT internal thread (only SS versions)                              |      |       |    |     |     |    |
| <b>Relief pressure</b>              |                                                                         |      |       |    |     |     |    |
| 10                                  | Preset relief pressure 10 bar                                           |      |       |    |     |     |    |
| 16                                  | Preset relief pressure 16 bar                                           |      |       |    |     |     |    |
| <b>Body and connection material</b> |                                                                         |      |       |    |     |     |    |
| PP                                  | Polypropylene                                                           |      |       |    |     |     |    |
| PVC                                 | Polyvinyl chloride                                                      |      |       |    |     |     |    |
| PV                                  | Polyvinylidene fluoride (PVDF)                                          |      |       |    |     |     |    |
| SS                                  | Stainless steel, grade 1.4571 according to EN 10027-2                   |      |       |    |     |     |    |
| <b>Gasket material</b>              |                                                                         |      |       |    |     |     |    |
| V,E                                 | FKM and EPDM gaskets are enclosed                                       |      |       |    |     |     |    |
| T                                   | PTFE gaskets are enclosed                                               |      |       |    |     |     |    |
| -                                   | No gaskets (only SS versions)                                           |      |       |    |     |     |    |
| <b>Hose or pipe connections</b>     |                                                                         |      |       |    |     |     |    |
| U2                                  | Hose, 4/6 mm, 6/9 mm, 6/12 mm, 9/12 mm                                  |      |       |    |     |     |    |
| U7                                  | Hose, 0.17" x 1/4"; 1/4" x 3/8"; 3/8" x 1/2"                            |      |       |    |     |     |    |
| A                                   | Threaded connections, Rp 1/4 internal thread                            |      |       |    |     |     |    |
| V                                   | Threaded connections, 1/4 NPT internal thread                           |      |       |    |     |     |    |
| X                                   | No connections included, please order separately from accessories range |      |       |    |     |     |    |

## Pressure relief valves from 60 to 460 l/h

| Code                                | Description                                                                                     | PRV- | G5/4- | 10 | PP/ | V,E | U3 |
|-------------------------------------|-------------------------------------------------------------------------------------------------|------|-------|----|-----|-----|----|
| PRV                                 | Pressure relief valve                                                                           |      |       |    |     |     |    |
| <b>Connection size</b>              |                                                                                                 |      |       |    |     |     |    |
| G5/4                                | G 5/4 external thread, groove for O-ring on top                                                 |      |       |    |     |     |    |
| G3/4                                | Rp 3/4 internal thread (only SS versions)                                                       |      |       |    |     |     |    |
| 3/4N                                | 3/4 NPT internal thread (only SS versions)                                                      |      |       |    |     |     |    |
| <b>Relief pressure</b>              |                                                                                                 |      |       |    |     |     |    |
| 10                                  | Preset relief pressure in bar                                                                   |      |       |    |     |     |    |
| <b>Body and connection material</b> |                                                                                                 |      |       |    |     |     |    |
| PP                                  | Polypropylene                                                                                   |      |       |    |     |     |    |
| PVC                                 | Polyvinyl chloride                                                                              |      |       |    |     |     |    |
| PV                                  | Polyvinylidene fluoride (PVDF)                                                                  |      |       |    |     |     |    |
| SS                                  | Stainless steel, grade 1.4571 according to EN 10027-2                                           |      |       |    |     |     |    |
| <b>Gasket material</b>              |                                                                                                 |      |       |    |     |     |    |
| V,E                                 | FKM and EPDM gaskets are enclosed                                                               |      |       |    |     |     |    |
| T                                   | Gasket material PTFE                                                                            |      |       |    |     |     |    |
| -                                   | No gaskets (only SS versions)                                                                   |      |       |    |     |     |    |
| <b>Hose or pipe connections</b>     |                                                                                                 |      |       |    |     |     |    |
| U3                                  | 2x Union nut, G 5/4<br>2x Hose connector, 19/20 mm<br>2x Hose clamp<br>2x Pipe connector, 25 mm |      |       |    |     |     |    |
| A7                                  | 2x Union nut, G 5/4<br>2x Inlay external thread 3/4 NPT                                         |      |       |    |     |     |    |
| A1                                  | Threaded connections, Rp 3/4 internal thread                                                    |      |       |    |     |     |    |
| A3                                  | Threaded connections, 3/4 NPT internal thread                                                   |      |       |    |     |     |    |
| X                                   | No connections included, please order separately from accessories range                         |      |       |    |     |     |    |

## 6. Maintaining the product

The product must only be serviced by authorised and qualified persons.

When working with chemicals, the accident prevention regulations applicable at the installation site must be applied.



Observe the chemical manufacturer's safety data sheets when handling chemicals.

When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.

Observe section [4.1 Handling the product](#).

Wipe up spilled liquid immediately to avoid slipping hazard.

### 6.1 Maintenance schedule

| Interval                                                                    | Task                                                                                                                                                     |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| After 8000 operating hours or at least every 12 months or in case of faults | If necessary, clean all product surfaces with a dry and clean cloth.<br>Replace the diaphragm. See section <a href="#">6.3 Replacing the diaphragm</a> . |

### 6.2 Cleaning

If necessary, clean all product surfaces with a dry and clean cloth.

### 6.3 Replacing the diaphragm

For service parts see service kit catalogue:

[http://net.grundfos.com/qr/i/96488862\\_23](http://net.grundfos.com/qr/i/96488862_23)

1. Shut down and drain the dosing system.
2. Make it impossible for a return flow or overpressure to occur.
3. Loosen the screws on the top part of the valve.
4. Remove the top part of the valve.
5. Remove the diaphragm.
6. Insert the new diaphragm.
7. Refit the top part of the valve and cross-tighten the screws with a torque wrench. Max. torque:

| Up to 60 l/h | 60-460 l/h | DN 65 |
|--------------|------------|-------|
| 2 Nm         | 4 Nm       | 5 Nm  |

8. Start up the dosing system again.
9. Re-tighten the screws on the top part of the valve after 48 operating hours with a torque wrench. Max. torque:

| Up to 60 l/h | 60-460 l/h | DN 65 |
|--------------|------------|-------|
| 2 Nm         | 4 Nm       | 5 Nm  |

## 7. Accessories

### 7.1 Union nut adapters (up to 460 l/h)

Union nut adapters allow direct assembly on the outlet valve of the dosing pump.

| Connections   | Material | Product number |
|---------------|----------|----------------|
| G 5/8 - G 5/8 | PVC      | 95730437       |
|               | PP       | 95730438       |
|               | PVDF     | 95730439       |
| G 5/8 - 1/4"  | SS*      | 96693273       |
|               | PP       | 99228667       |
| G 5/4 - G 5/4 | PVC      | 99228665       |
|               | PVDF     | 99228669       |

\* Stainless steel 1.4571

### 7.2 Set of counter flanges for pressure relief valves DN 65

| Size  | Material | Description                                                    | Product number |
|-------|----------|----------------------------------------------------------------|----------------|
| DN 65 | PVC      | Set includes lapped flange, headed bush, screws, washers, nuts | 96727602       |
| DN 65 | PP       |                                                                | 96727603       |
| DN 65 | SS*      | Set includes welding neck flange, screws, washers, nuts        | 96727604       |

\* Stainless steel 1.4571

## 8. Fault finding

| Fault                                                      | Possible cause                                                                | Possible remedy                                                                                                   |
|------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Medium is flowing out of the overflow outlet continuously. | Relief pressure is set too low.                                               | Set the valve to a higher relief pressure.                                                                        |
|                                                            | Only valves with T-piece installation: Valve is installed in wrong direction. | Remove the valve and install it in flow direction. The flow direction is indicated by an arrow on the valve body. |
|                                                            | Defective diaphragm.                                                          | Replace the diaphragm.                                                                                            |
|                                                            | Dirt in the pressure relief valve.                                            | Rinse the pressure relief valve.                                                                                  |
| The system pressure exceeds the expected maximum pressure. | Relief pressure is set too high.                                              | Set the valve to a lower relief pressure.                                                                         |
|                                                            | Only valves with T-piece installation: Valve is installed in wrong direction. | Remove the valve and install it in flow direction. The flow direction is indicated by an arrow on the valve body. |

## 9. Technical data

### 9.1 Permissible media temperature

| Body material | Maximum operating pressure |                  |
|---------------|----------------------------|------------------|
|               | up to 10 bar               | > 10 to 16 bar   |
| PVC, PP       | 0 °C to +40 °C             | 0 °C to +20 °C   |
| PVDF, SS      | -10 °C to +40 °C           | -10 °C to +20 °C |

### 9.2 Storage and ambient temperatures

|                                    | Maximum operating pressure |                |
|------------------------------------|----------------------------|----------------|
|                                    | up to 10 bar               | > 10 to 16 bar |
| Min./Max. ambient temperature [°C] | 0/+40                      | 0/+20          |
| Min./Max. storage temperature [°C] | -10/+50                    |                |

### 9.3 Pressure relief valves up to 60 l/h

#### 9.3.1 Technical data

|                               |       |                |
|-------------------------------|-------|----------------|
| Maximum operating pressure    | [bar] | 16             |
| Factory-set relief pressure   | [bar] | ~10<br>~16     |
| Relief pressure setting range | [bar] | ~5-10<br>~7-16 |
| Maximum flow rate             | [l/h] | 60             |

#### 9.3.2 Materials

| Materials |             |          |                                            | Figure |
|-----------|-------------|----------|--------------------------------------------|--------|
| Body      | Connections | Gaskets  | Type designation                           |        |
| PP        | PP          | FKM/EPDM | PRV-G5/8-10 PP/V,E<br>PRV-G5/8-16 PP/V,E   | 5      |
| PVC       | PVC         | FKM/EPDM | PRV-G5/8-10 PVC/V,E<br>PRV-G5/8-16 PVC/V,E |        |
|           |             | PTFE     | PRV-G5/8-10 PVC/T<br>PRV-G5/8-16 PVC/T     |        |
| PVDF      | PVDF        | FKM/EPDM | PRV-G5/8-10 PV/V,E<br>PRV-G5/8-16 PV/V,E   |        |
|           |             | PTFE     | PRV-G5/8-10 PV/T<br>PRV-G5/8-16 PV/T       | 6      |
| SS*       | -           | -        | PRV-G1/4-10 SS/-<br>PRV-G1/4-16 SS/-       |        |
| SS*       | -           | -        | PRV-1/4N-10 SS/-<br>PRV-1/4N-16 SS/-       |        |

\* Stainless steel 1.4571

### 9.3.3 Dimensional drawings

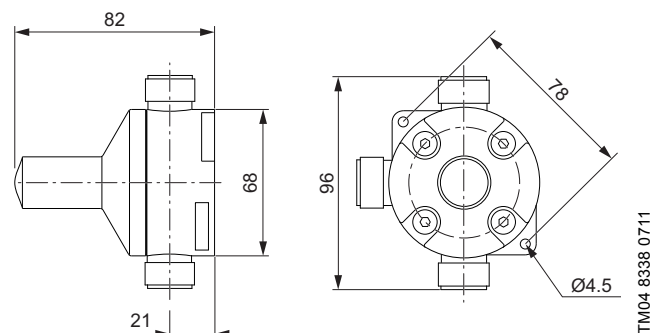


Fig. 5 Dimensions of PP, PVC, PVDF valves

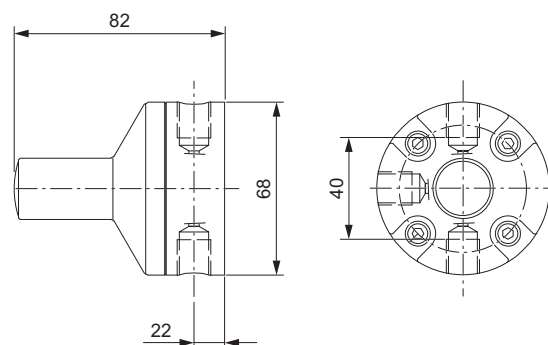


Fig. 6 Dimensions of SS valves



## 9.4 Pressure relief valves from 60 to 460 l/h

### 9.4.1 Technical data

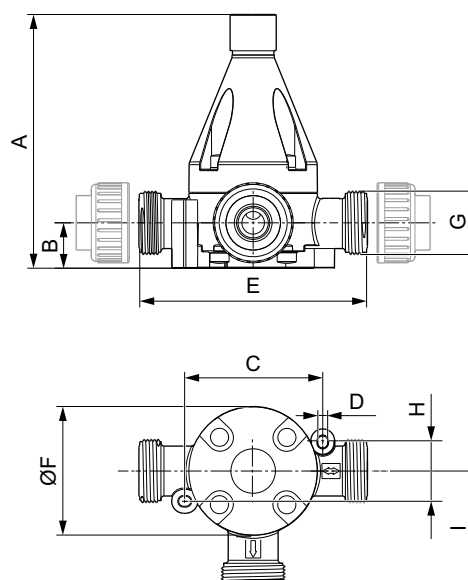
|                               |       |       |
|-------------------------------|-------|-------|
| Maximum operating pressure    | [bar] | 10    |
| Factory-set relief pressure   | [bar] | ~10   |
| Relief pressure setting range | [bar] | ~3-10 |
| Maximum flow rate             | [l/h] | 460   |

### 9.4.2 Materials

| Materials |             |          |                     | Figure |
|-----------|-------------|----------|---------------------|--------|
| Body      | Connections | Gaskets  | Type designation    |        |
| PP        | PP          | FKM/EPDM | PRV-G5/4-10 PP/V,E  | 7      |
| PVC       | PVC         | FKM/EPDM | PRV-G5/4-10 PVC/V,E |        |
|           |             | PTFE     | PRV-G5/4-10 PVC/T   |        |
| PVDF      | PVDF        | FKM/EPDM | PRV-G5/4-10 PV/V,E  |        |
|           |             | PTFE     | PRV-G5/4-10 PV/T    | 8      |
| SS*       | -           | -        | PRV-G3/4-10 SS/-    |        |
| SS*       | -           | -        | PRV-3/4N-10 SS/-    |        |

\* Stainless steel 1.4571

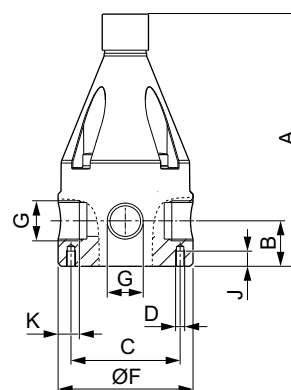
### 9.4.3 Dimensional drawings



TM06 8513 0817

Fig. 7 Dimensions of PP, PVC, PVDF valves

| A    | B    | C    | D    | E    | ØF   | G     | H    | I    |
|------|------|------|------|------|------|-------|------|------|
| [mm] | [mm] | [mm] | [mm] | [mm] | [mm] |       | [mm] | [mm] |
| 168  | 30   | 92   | 6.5  | 150  | 85   | G 5/4 | 40   | 75   |



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Fig. 8 Dimensions of SS valves

| A    | B    | C    | D   | ØF   | G      | J    | K    |
|------|------|------|-----|------|--------|------|------|
| [mm] | [mm] | [mm] |     | [mm] |        | [mm] | [mm] |
| 167  | 30   | 63   | M 6 | 89   | Rp 3/4 | 10   | 17.5 |

## 9.5 Pressure relief valves DN 65

### 9.5.1 Technical data

|                            |       |      |
|----------------------------|-------|------|
| Maximum operating pressure | [bar] | 10   |
| Maximum flow rate          | [l/h] | 4000 |

| Factory-set relief pressure [bar] | Relief pressure setting range [bar] | Nominal width [mm] | Body | Gasket    | L [mm] | Connection   | Product number | Figure |
|-----------------------------------|-------------------------------------|--------------------|------|-----------|--------|--------------|----------------|--------|
| ~3                                | ~0.5 - 6                            | 65                 | PVC  | FKM       | 266    | Flange DN 65 | 96638461       | 9      |
|                                   |                                     | 65                 | PVC  | EPDM      | 266    | Flange DN 65 | 96635243       |        |
| ~4                                | ~1-10                               | 65                 | SS*  | PVC, soft | 394    | Flange DN 65 | 96694452       | 10     |

\* Stainless steel 1.4571

### 9.5.2 Dimensional drawings

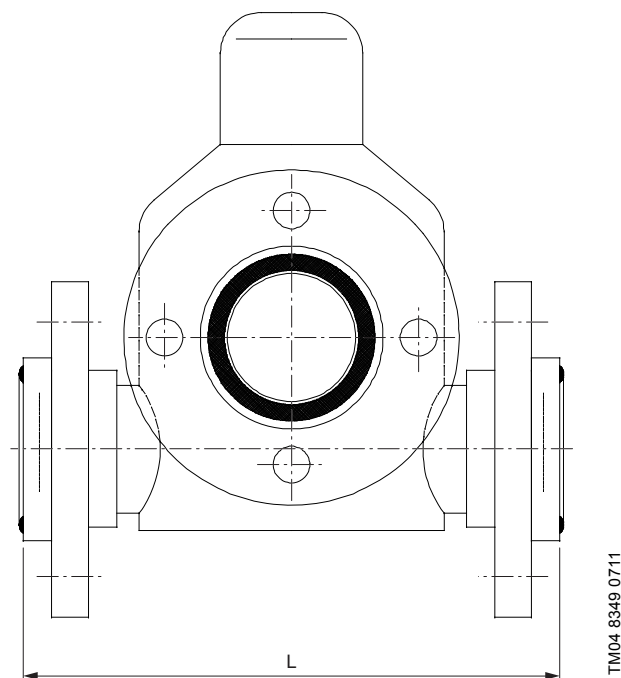


Fig. 9 Dimensions of PPH, PVC, PP valves

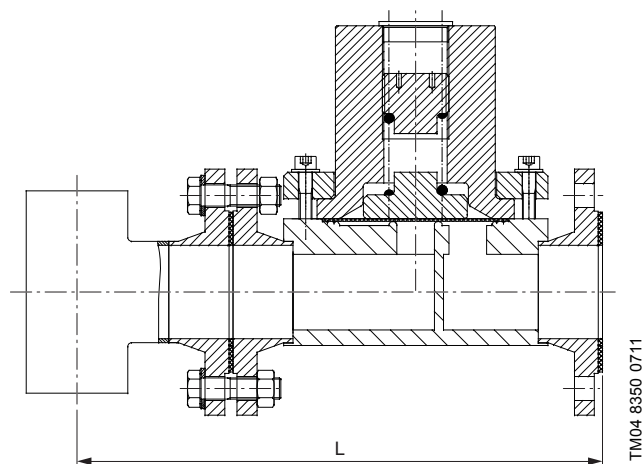


Fig. 10 Dimensions of SS valves with flange

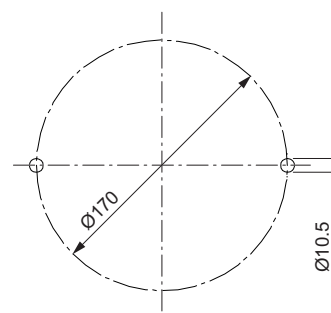


Fig. 11 Drilling scheme for SS valves

## 10. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

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